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# **Project Title**

**Enhancing workforce management at Infor Relay Systems  
' (Pvt) Ltd through implementation of a Real-Time RFID  
employee attendance system.**

# Problem Statement

At Infor Relay Systems (Pvt) Ltd, recording attendance on time and correctly is very important for smooth operations and good service delivery. However, many employees record their attendance late. This causes problems such as incorrect attendance records, payroll errors, poor planning, and reduced productivity. Therefore, there is a need to improve the attendance recording process to ensure it is done on time and accurately.

# Aims and Objectives

**Aim:** To develop a reliable and efficient RFID employee attendance system that enhances the accuracy of attendance tracking and reduces errors in the payroll process.

## **Objectives:**

1. Automating attendance tracking.
2. Ensuring accurate labour costing.
3. Integrating RFID attendance with existing systems.

# Methodology

Infor Relay Systems (Pvt) Ltd relies on manual attendance registers and supervisor verification to monitor employee attendance. This traditional system has resulted in:

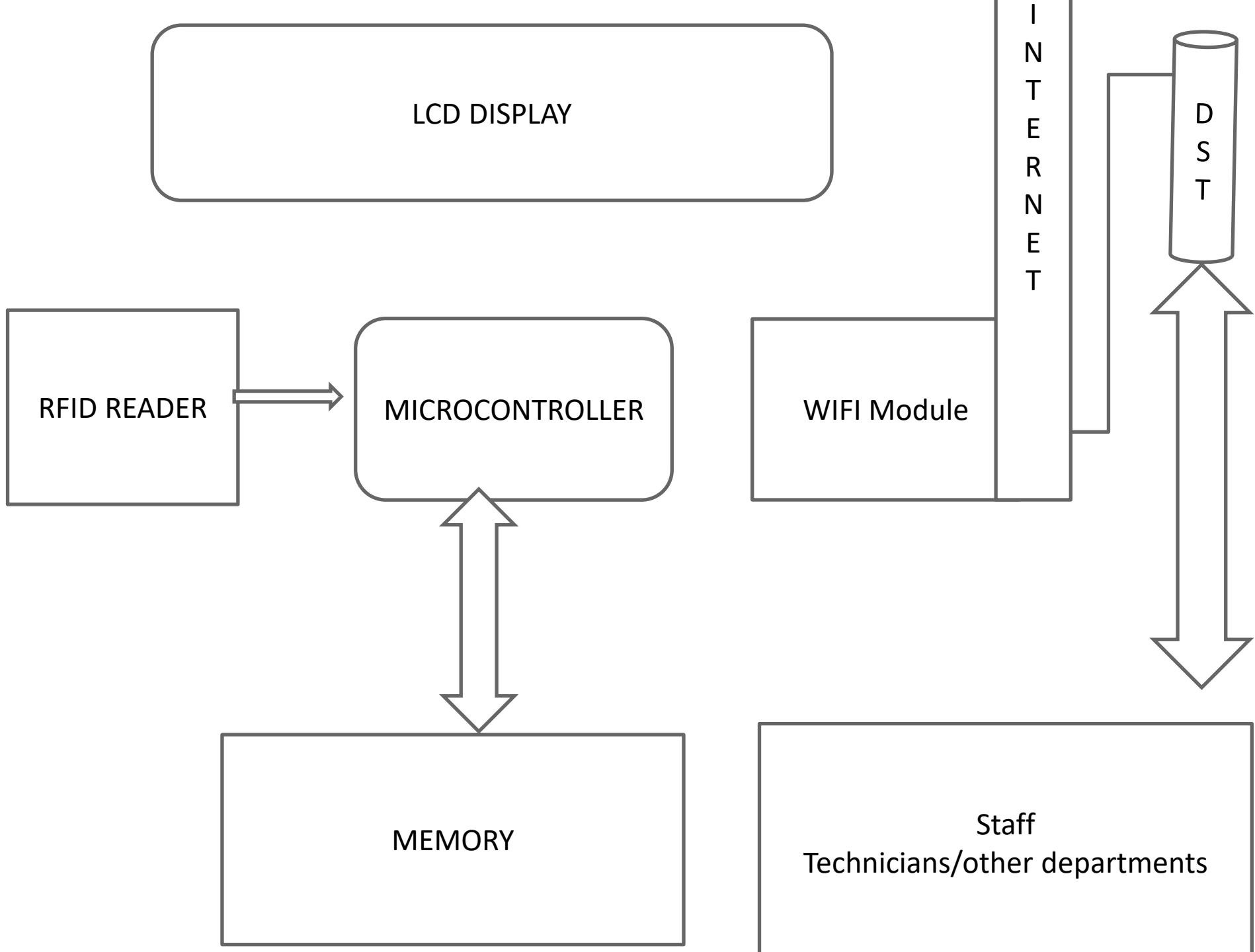
Inaccurate attendance records

Time theft

Delays in payroll processing

Poor workforce accountability

To improve efficiency and transparency, a Real-Time RFID-based employee attendance system is proposed.



# Technology Deployment

## Installation of Hardware

The following components were deployed:

1. RFID Reader at main entrance
2. RFID cards issued to all employees
3. ESP32/Arduino microcontroller
4. WiFi/Ethernet connectivity

## System Configuration

Unique ID assigned to each employee

Database created for employee records

Attendance logging programmed with real-time timestamp

Web dashboard developed for HR access

# Automation and Security

The system automatically

1. Records entry and exit time
2. Updates database instantly
3. Flags duplicate or invalid scans
4. Calculates total working hours

Security

Password-protected dashboard

Role-based access control (Admin, HR, Manager)

Encrypted data transmission (HTTPS)

Unique RFID tag authentication



# Evaluation and Scale-up

## Performance Evaluation

After implementation, results showed:

Reduction in attendance fraud

Improved payroll accuracy

Faster report generation

Reduced administrative workload

Real-time employee tracking